

PART C: Presentation and Justification of Solution Design Requirements

Based on research into existing tree stand solutions and hands-on testing of the initial prototype, several core design requirements were identified to guide the development of an improved, lightweight, and functional 3D-printed tree stand. These criteria were developed to ensure that the final product meets the practical needs of hunters while maintaining safety and durability in the field.

The Stand Must Be Safe and Structurally Reliable

Safety is the highest priority for any elevated hunting platform. The stand must support the weight of an average adult hunter while maintaining stability and resisting environmental stress such as wind, vibration, and shifting movement. During prototype testing, a slight flex in the material was observed, which, while not structurally compromising, signaled the importance of reinforcing critical load-bearing zones or selecting stronger composite materials in future versions.

The Stand Must Be Lightweight for Easy Transport

A primary objective of this design is to reduce the physical strain of carrying a traditional tree stand. Hunters often trek long distances through challenging terrain, so minimizing weight significantly improves mobility and overall experience. The use of 3D-printed polymer materials allowed the prototype to shed several pounds compared to metal counterparts, proving that substantial weight savings are achievable.

The Design Must Be Quiet and Stealthy in Use

Metal-on-metal noise during setup or shifting movement can easily alert nearby game. The tree stand must be designed with low-noise operation in mind. The use of plastic and composite materials provides a natural dampening effect, reducing unwanted sound during setup and while in the tree.

The Stand Must Be Durable Enough for Repeated Outdoor Use

While lightweight and quiet, the materials and design must also be rugged enough to endure repeated use in outdoor environments. Exposure to moisture, UV rays, and temperature changes must be considered in material selection. Reinforcement strategies or protective coatings may be necessary to increase long-term performance without adding significant weight.

The Solution Must Be Cost-Effective to Produce or Replicate

One of the advantages of additive manufacturing is its cost efficiency in low-volume production. This design should leverage the flexibility of 3D printing to remain affordable for hunters who want a customized solution without the premium price tag of ultralight commercial models. The

prototype's simple part geometry was intentionally designed for easy replication and minimal print time to keep material and production costs low.

The Design Should Be Modular or Easy to Disassemble

Hunters often need to pack and move gear between hunting sites. A good stand should be easy to break down, transport, and reassemble. The design should avoid complex hardware and allow for tool-free or quick assembly in the field, further supporting mobile hunting methods.

After testing the prototype and reviewing feedback from field users, these design requirements were confirmed as critical for a successful lightweight tree stand. Particular attention was given to material flex and structural response, which, while acceptable in the current model, will guide adjustments in the final design for improved performance and safety assurance.